1. How does the bracelet detect removal from the child's wrist?

The system monitors presence through the MQTT Last Will mechanism. The bracelet publishes a status heartbeat on every wake cycle, if the broker does not receive a check-in within the configured interval plus the few minute grace period, it fires the Last Will and the dashboard raises an alert showing the device name, last battery level, and timestamp of last contact. This tells you a bracelet has gone offline. A dedicated **wrist-removal sensor** is not in the **current design**, if distinguishing physical removal from other offline causes is a hard requirement, that is a decision we need to make before the schematic design is finalized as it affects the PCB and enclosure.

2. Where is OTA firmware updating in the architecture?

OTA was not explicitly mentioned in the document but is always considered for devices that can communicate wirelessy with the internet, and will be implemented in Phase 4 alongside the core firmware. The **ESP32-S3 module** supports native over-the-air updates via a dual-partition flash scheme, the device downloads new firmware in the background while operating normally, validates it, and switches partitions on the next restart. Failed updates automatically roll back to the previously validated firmware. No physical access to any bracelet is required.

3. What child safety certifications will be required?

The **ESP32-S3-WROOM-1-N8R8** module already carries FCC Part 15 and CE radio certifications, which transfer to the end product and cover RF emissions compliance. For the complete product as a child wearable, EN 71 in Europe or ASTM F963 in the United States covers physical safety materials, mechanical integrity, and small parts. RoHS compliance applies across all markets. These are post-prototype activities and I will provide the full technical documentation package a certification laboratory will need.

4. How is user authentication handled?

Authentication operates at two independent layers.

The first is dashboard authentication. Caregivers log in with email and password. The backend issues signed JWT tokens on successful login, every subsequent request carries this token and is validated before any action is processed. New accounts require explicit administrator approval before they can authenticate. Revocation is immediate when user account needs to be terminated from the **ADMIN ACCOUNT.**

The second is MQTT broker authentication. Each bracelet holds unique credentials, client ID, username, and passwords hard-coded at flash time and registered on HiveMQ before deployment. The dashboard backend holds its own separate publisher credentials stored only in the server environment, never exposed to any user. The broker enforces an Access Control List restricting each set of credentials to its own topic scope. These two layers are independent, compromising one does not

expose the other.

5. What data is stored in the cloud?

Permanently — removing these breaks the system: Per-device MQTT credentials on the broker and caregiver account data on the backend database (The caregiver account data is simply the information that allows a caregiver to log into the dashboard their email address, their hashed password, their account status (active or pending), and their role (administrator or standard caregiver). Without these persisting in the database, the login system has nothing to validate against and authentication breaks entirely)

Temporarily — discarded after confirmed delivery: Message images are uploaded to cloud storage and deleted automatically once the bracelet publishes a delivery acknowledgment confirming the image was displayed. The MQTT retained message is cleared at the same point. In deep sleep mode the maximum time an image sits in cloud storage is 5 minutes. In modem sleep it is seconds. Nothing persists beyond confirmed delivery.

Configurable: Delivery logs — which device displayed which message at what time — and battery status history can be retained for a defined window for operational auditing. If historical query-ability is not needed, these can be discarded after each session. Default recommendation is 30 days.

6. What happens when a bracelet is lost or reassigned?

Lost: The administrator marks the device inactive on the dashboard. If permanent loss is confirmed, the device's broker credentials are revoked, it cannot reconnect or receive messages even if recovered. If later found, credentials are reinstated with no hardware intervention needed.

Reassigned: A long press on the recessed reset button clears all provisioning data from the device, device name, Wi-Fi credentials, and message history. The bracelet reboots into setup mode for re-provisioning with the new child's name. Broker credentials on the device remain unchanged, only the human-readable name updates. The dashboard reflects the change on the next check-in.

7. What is the enclosure size and estimated weight?

Final mechanical dimensions is based on the locked hardware which is the display panel at approximately 40mm × 32mm (https://www.buydisplay.com/1-69-inch-240x280-round-rectangle-ips-tft-lcd-screen-connector-fpc), 12–14mm thickness.

8. What is the worst-case battery life?

Worst case combines maximum message frequency with a battery at end-of-life capacity, lithium polymer degrades to approximately 80% after 12 months of daily charging, compounding the 20% de-rating already applied in the power analysis.

Battery life is also greatly influenced by message frequency and interrupts in the resting phase of the

wearable
At 2 messages per hour on an end-of-life cell: deep sleep mode delivers approximately 24 hours (**theoretical expected value**) well above the 10-hour minimum. Modem sleep delivers approximately 11 hours, making daily charging operationally mandatory as the battery ages.

9. How does the system scale beyond 20 bracelets?

The 20-device figure reflects the consumer Wi-Fi router constraint and HiveMQ connection tier, not any limitation in the bracelet hardware, firmware, or backend. Wake time randomization is already built into the design to prevent connection storms at any device count.

Scaling beyond 20 requires upgrading the access point (**Router**) to a business-grade alternative, Ubiquiti UniFi or TP-Link Omada are appropriate for a facility environment, and upgrading the HiveMQ broker tier, which supports up to 100 concurrent connections on its entry paid plan. No changes to bracelet hardware or firmware are required to scale.

10. What security prevents unauthorised users from sending messages?

Three independent layers enforce the security chain in sequence.

The dashboard requires a valid authenticated session before any API request is processed, no approved account, no access, no request reaches the broker.

The MQTT broker requires valid credentials on every connection attempt. Publisher credentials exist only in the backend server environment. The broker ACL restricts each credential set to defined topic permissions, a compromised bracelet credential affects only that device and cannot interact with any other.

All connections, frontend to backend, backend to broker, and bracelet to broker, operate over TLS 1.2 on port 8883. Intercepted traffic is fully encrypted. No single point of failure exists across the three layers.